

YUEH-HSUAN (Helena) HUANG

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EDUCATION

National Taiwan University (NTU), Taipei, Taiwan

Bachelor of Science in Electrical Engineering

Sep 2022 – Present

- Overall GPA: 4.12/4.30
- Program: Business Analytics Program, College of Management

INDUSTRY EXPERIENCE

AI Intern, Future Technology Department, TVBS Media Inc., Taipei, Taiwan

– Project Title: AI-Powered News Scheduling Automation

Jul 2025 – Present

- Developed an end-to-end 6 am/12 am news automation workflow that decreased editorial workload by 75%.
- Leveraged n8n and the Google Gemini API to transform a 20-year veteran editor's expertise into a multi-modal AI scoring system for 4000+ articles.
- Engineered a solution on Google Cloud Run to scrape Google Trends data efficiently, boosting data-driven hot-topic coverage by over 10x and replacing inefficient RSS-driven methods.

RESEARCH EXPERIENCE

Interdisciplinary Data Science & Signal Processing Lab, NTU, Taipei, Taiwan

– Undergraduate Researcher, Advisor: Prof. Che Lin

Sep 2024 – Present

Topic: Parameter-Efficient Transfer Learning for Multi-Modal Sequential Recommendation

- Investigated PEFT methods (LoRA, Adapters) to fine-tune a state-of-the-art ICML-published multi-modal recommender model under funding from the National Science and Technology Council (NSTC).
- Created robust preprocessing pipelines across multi-domain datasets to support transfer learning baseline experiments.

Digital Image and Signal Processing Lab, NTU, Taipei, Taiwan

– Undergraduate Researcher, Advisor: Prof. Jiang-Jiun Ding

Sep 2024 – Present

Topic: Human Face Recognition and Character Verification Using Image Processing Techniques

- Built a multi-stage face detection pipeline incorporating skin-tone classification via SVM, facial landmark detection, and geometric consistency verification, achieving a 95.5% F1-score with 1.0 precision.
- Implemented a handwriting character verification system using zonal pixel density and stroke geometry, reaching 88% classification accuracy on a 1,000-sample dataset.

Institute of Physics, Academia Sinica, Taipei, Taiwan

– Undergraduate Research Intern, Advisor: Dr. Hong-Yan Shih

Jul 2024 – Aug 2024

Topic: Tensor Networks for Interpretable Supervised Learning Tasks

- Investigated the use of Tensor Networks for interpretable machine learning, and reproduced a Matrix Product State model trained via a Density Matrix Renormalization Group inspired algorithm.
- Achieved 98% accuracy on a proof-of-concept task and presented findings at the institute-wide poster session.
- Analyzed model complexity, discovering that model parameter counts grow approximately linearly with the square of vector dimensions.

PROJECT EXPERIENCE

AI-Powered Product Compliance System for a Leading E-commerce Platform

– Prog5265 Capstone for Business Analytics, in collaboration with momo.com Inc.

Feb 2025 – Jun 2025

- Led a team of 3 economics students to deliver a multi-modal violation detection system for 4,700+ e-commerce products, achieving 5x improvement over official methods using LLMs and Google Cloud OCR
- Prototyped an automated end-to-end workflow using n8n, cutting processing time by over 83% and establishing a scalable framework for future applications.

An Automated Alarm System with Sleep Detection Implementation

–EE4057 Electrical Engineering Lab (Biomedical Engineering)

Oct 2024 – Dec 2024

- Co-designed a sleep onset detector using EEG data analysis.
- Proposed a preprocessing pipeline with bandpass filtering and Morlet wavelet convolution for spindle detection.

"Liver! So Tired" - A 2D Game in C++/SDL

– EE1004 Computer Programming

Oct 2022 – Dec 2022

- Co-developed a 2D projection-based game inspired by Angry Birds using OOP principles and friend-class collision detection.

PUBLICATIONS

[1] Luo, Y.-X., Lin, Y.-C., Chuang, M.-T., Chen, J.-H., Tsai, I.-N., Kiew, P.X., **Huang, Y.-H.**, Liu, C.-F., Chen, Y.-C., Feng, B.-H., Ren, W., Lee, H.-y. (2025) ToxicTone: A Mandarin Audio Dataset Annotated for Toxicity and Toxic Utterance Tonality. Proc. **Interspeech 2025**, 4008-4012, doi: 10.21437/Interspeech.2025-679

[2] Tsai, I.-N., Wu, Y.-X., **Huang, Y.-H.**, Chen, Y.-C., & Ding, J.-J. (2025). Optimization of Elevator Standby Scheduling Strategy in Smart Buildings. **Applied System Innovation**, 8(5), 132. <https://doi.org/10.3390/asi8050132>

[3] Lin, Y. C., Chen, Y. H., Dong, J. K., **Huang, Y. H.**, Chen, S. C., Chen, Y. C., ... & Lee, H. Y. (2025). TAU: A Benchmark for Cultural Sound Understanding Beyond Semantics. arXiv preprint arXiv:2509.26329. **(Under Review)**

TECHNICAL SKILLS

- Programming Languages: Python, C++, JavaScript, MATLAB
- Cloud & DevOps: Google Cloud Platform, Git/Git Actions, n8n
- Tools: Elasticsearch, n8n (Workflow Automation), API Integration, BMad Method
- Machine Learning & Data Science: Pytorch, Image & Signal Processing, Recommendation Systems

LEADERSHIP & EXTRACURRICULARS

Vice-Captain, NTUEE Women's Basketball Team

Sep 2022 – Jun 2023; Sep 2024 – Present

- Contributed to team's championship victories in both the 2025 NTU Cup and National EE Cup as the sixth man.
- Supported team leadership through consistent attendance and maintaining team morale.
- Managed all team logistics and finances, securing 130+ practice venues annually through negotiation with other teams and organizing 10+ practice matches annually to build team chemistry.